

Vikas Zende

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Professional Summary

Full Stack Developer / MERN Stack Developer with hands-on experience building real-time, user-centric applications serving 100+ users. Skilled in the complete development lifecycle, from designing responsive UIs to deploying secure and scalable backends, achieving up to 40% improvements in performance and accessibility. Pursuing Honors in AI/ML with exposure to integrating intelligent features into web solutions. Passionate about writing clean, maintainable code and continuously growing through internships and personal projects.

Skills

Languages: JavaScript, Python, SQL.

Frameworks/Libraries: React.js, Next.js, Node.js, Express.js, FastAPI, Flask, Tailwind CSS

AI/ML: NumPy, Pandas, Pydantic, LangChain, LLM Integration

Databases: MySQL, MongoDB, PostgreSQL

Tools: Git, GitHub, Postman, REST APIs, WebSocket, JWT, Railway, Vercel.

Technical: Full Stack Development, API Integration, Responsive Design, Prompt Engineering, Data Processing, CI/CD

Soft Skills: Communication, Teamwork, Problem-Solving, Adaptability, Attention to Detail-

Work Experience

Frontend Web Developer Intern

May 2024 – June 2024

K.J. Somaiya College of Engineering, Mumbai

- Built a mobile-first customer feedback system using HTML5, CSS3, and JavaScript.
- Designed a responsive UI with form validation to collect and manage customer feedback.
- Ensured cross-browser compatibility and accurate data capture.

Full Stack Web Developer & PLC Automation Intern

Dec 2024 – Jan 2025

K.J. Somaiya College of Engineering, Mumbai

- Developed a Student Attendance Tracker using Next.js, MySQL, Tailwind CSS, and shadcn/ui.
- Implemented RESTful APIs with Kinde authentication for secure user access.
- Contributed to PLC automation by developing an HMI-based Four-Way Traffic Light Control system.

Projects

Hybrid RAG Platform

GitHub

Personal Project

- Built an AI-powered Retrieval-Augmented Generation (RAG) platform using FastAPI, PostgreSQL, Redis, pgvector, and Groq LLM.
- Implemented hybrid search with BM25, vector search, reranking, semantic caching, and citation-based responses.
- Developed a document ingestion pipeline for PDF processing, chunking, embeddings, and conversational AI.

Scalable URL Shortener API

Demo

Personal Project

- Developed a scalable URL shortener using Node.js, Express.js, PostgreSQL, and Redis with a layered architecture.
- Implemented JWT authentication, custom short URLs, click analytics, Redis caching, rate limiting, and REST APIs.
- Optimized redirects using Redis caching for scalability, low latency, and high availability.

Industry 4.0 – Automation & Fabrication

May 2025 – May 2026

Academic Major Project

- Built a YOLOv11s-based defect detection system on a 400-image dataset, achieving 98.2% precision and 100% recall.
- Developed an ESP32-based conveyor, servo sorting system, and robotic arm with 91% real-time classification accuracy.
- Designed 3D-printed packaging and automated handling workflows, reducing manual work by approximately 90%.

Education

Bachelor of Engineering – Electronics and Telecommunication (Honors in AI/ML)

Rajiv Gandhi Institute of Technology, University of Mumbai

CGPA: 8.89 / 10

Graduated: 2026